

Theodoros Giannakopoulos

GENERAL INFO	Date of birth09/04/1980 Place of birthAthens, Greece E-mailtyiannak@gmail.com Cell(+30) 6977157480 WWWtyiannak.github.io Google Scholaruser=BeIoqhwAAAAJ Githubgithub.com/tyiannak Mediummedium.com/@tyiannak
EDUCATION	PHD, University of Athens, Dept of Informatics and Telecommunications, 2009 - Thesis Title: <i>Study and application of acoustic information for the detection of harmful content, and fusion with visual information</i>, Advisor: Professor Sergios Theodoridis MSC, Signal and Image Processing Systems, 2004 - With Honors (final grade: 9.34/10) - Thesis Title: <i>Design and implementation of real-time processing methods of speech signals, using microphone arrays</i>, Advisor: Professor Nikos Fakotakis B.S., University of Athens, Dept of Informatics and Telecommunications, 2002 - Final grade: 7.14/10 - Thesis Title: <i>Design and implementation of a semi-automatic method for boundary detection in high resolution ultrasound images</i>
ACHIEVEMENTS AND AWARDS	Over 100 publications, over 2600 citations, 24 h-index (gscholar) 9 open source Python github repos. Top 0.1% rank (over 0.6M Python users) Postdoc scholarship (April 2017 - April 2019) funded by the Greek State Scholarships Foundation (IKY). Subject: "Soundscape quality analysis using machine learning", Participant: CIL, IIT, NCSR Demokritos, Greece Winner of the IEEE International Workshop on Multimedia Signal Processing (MMSP07) student paper contest, paper "A multi-class audio classification method with respect to violent content in movies, using bayesian networks" 1st award in 9th International Conference on Machine Learning and Applications Speaker Clustering Challenge (2010), paper "Unsupervised Speaker Clustering in a Linear Discriminant Subspace". 2nd Position award in The 2nd e-Prevention challenge: Psychotic and Non-Psychotic Relapse Detection using Wearable-Based Digital Phenotyping of ICASSP 2024, paper "A self-supervised learning approach for detecting non-psychotic relapses using wearable-based digital phenotyping".

Best-paper award nomination in 12th International Conference on Affective Computing and Intelligent Interaction, paper “Emotion-Aware Speech Popularity Prediction: A Use-Case on Ted Talks”

PROFESSIONAL
EXPERIENCE

OCTOBER 2020 - TODAY: Principal researcher at the National Center for Scientific Research “Demokritos”, Greece.

SEPTEMBER 2017 - TODAY: Director of Machine Learning, at Behavioral Signals, LA, USA

MARCH 2017 - AUGUST 2017: Machine Learning Engineer at Soundsnap.com, NY, USA

JUNE 2012 - SEPTEMBER 2017 Post-doc research member in the National Center for Scientific Research “Demokritos”, Greece. Research Projects:

- January 2016 - September: 2017 EU funded research project “Radio: Unobtrusive, Efficient, Reliable And Modular Solutions For Independent Ageing”
- August 2014 - September 2015: EU funded research project “Sentimagi: Brand monitoring and reputation management via multi-modal sentiment analysis” (2935)
- August 2013 - May 2015: EU funded research project “AMINESS: Analysis of Marine INformation for Environmentally Safe Shipping”
- June 2012 - February 2014: EU funded research project “USEFIL: Unobstrusive Smart Enviroments For Independent Living”

FEBRUARY 2012 - NOVEMBER 2015: Post-doc research member in the Institute for the Management of Information Systems (IMIS) of the Research and Innovation Centre in Information, Communication and Knowledge Technologies “Athena”, Greece

MAY 2011 - JANUARY 2012 Serving the Greek Army

JULY 2009 - APRIL 2011: Post-doc research member in the EU funded research project “Computer-Aided Semantic Annotation of Multimedia - CASAM”. Participant: CIL, IIT, NCSR Demokritos, Greece

FEBRUARY 2009 - JUNE 2009: Research member in the EU funded research project “Prometheus” (PRediction and interpretation of huMan bEhaviour based on probabilistic sTructures and HEterogeneoUs Sensors - FP7-214901). Participant: University of Patras, The Wire Communications Laboratory.

DECEMBER 2005 - NOVEMBER 2008: Postgraduate research member in the research project “PENED - 03ED375”. Participant: University of Athens, Department of Informatics and Telecommunications.

DECEMBER 2004 - NOVEMBER 2005: Postgraduate research member in the IST I-IMAS project (Intelligent Imaging Sensors for Industry, Health and Security). Participant: Computer Technology Institute and Press “Diophantus”

FEBRUARY 2003 - SEPTEMBER 2004: Postgraduate research member in the European Union IST research projects INSIPRE (INfotainment management with Speech Interaction via REMote-microphones and telephone interfaces, IST-2001-32746) and 2023 ITEA-AMBIENCE: Context aware environments for ambient services. Participant: University of Patras, The Wire Communications Laboratory.

MARCH - APRIL 2002: Intern researcher in the Institute of Child Health, University of Birmingham. Subject: Telemedicine and Medical Image Processing.

TEACHING EXPERIENCE

IIT, NCSR Demokritos:

Teaching in Master Programs:

- 1 MSc of Data Science, NCSR Demokritos
 - 1 2018 - 2020: "Multimodal Information Processing and Analysis"
 - 2 2021 - today: "Deep Learning"
 - 3 2021 - today: "Data Programming"
- 2 MSc of Artificial Intelligence, NCSR Demokritos
 - 1 2021 - today: "Machine Learning"
 - 2 2021 - today: "Deep Learning"
 - 3 2021 - today: "Multimodal Information Processing and Analysis"

Diploma and Master Thesis (co)supervision:

- 1 Over 20 Undergraduate theses and over 60 Postgraduate Theses
- 2 During the candidate's tenure as a Researcher Grade B: 6 Supervision of Undergraduate Theses and 27 Supervision of Postgraduate Theses.

PhD (co)supervision:

- 1 Person Search Using Identity Attributes, Nikolaos Sarafianos, 2015 - 2019, University of Houston, Texas, USA
- 2 From body to brain: using artificial intelligence to identify user skills and intentions in interactive scenarios, Michalis Papakostas, 2015 - 2019, University of Texas at Arlington, USA

Technological Educational Institute (T.E.I) of Lamia

Scientific associate, Courses: Information Theory, Data Structures. 2016-2018 (3 Semesters)

University of Athens, Dept of Informatics and Telecommunications

Teaching and laboratory assistant for the undergraduate course: *Introduction to Digital Signal Processing*, 2004 TO 2008

University of Patras, Dept Computer Engineering and Informatics Department

Teaching and laboratory assistant for the undergraduate course *Introduction to Programming*, 2002 TO 2004

RESEARCH ACADEMIC PROJECTS [14 TOTAL]

NCSR DEMOKRITOS - AS TENURED RESEARCHER PI OR TECHNICAL MANAGER [7 TOTAL, SEE TABLE 1]

Also, actively participated in the following projects [8 Total]: Deploy AI, Pharos, AI4Europe, AI4Trust, MiTrap, SmartAttica, SmartHealth and AI4Copernicus.

Project Name	Role	Funding	Period	NCSR Budget
MSc AI	PI	Educational	2022-today	Variable
MSc Data Science	PI	Educational	2022-today	Variable
Museek	PI	National	2022-2024	195
Behavioral Signals	PI	Private	2022-today	70 p year
Smartsubs	Technical Manager	National	2021-2023	262K
Faradai	PI	EU (EDF)	2023-today	600K

Table 1: Projects List Managed as Tenured Researcher (2000-)

NCSR DEMOKRITOS - As POSTDOC[6 TOTAL]: CASAM-11488, USEFIL-11669, AMINESS-11798, SentIMAGi-11883, Radio 11918, Post-doc IKY Scholarship (2017)

ATHENA [3 TOTAL]: ESPAS, INTERACT, OPENAIRE2020

NATIONAL AND KAPODESTRIAN UNIVERSITY OF ATHENS [1 TOTAL]: PENED-03698

UNIVERSITY OF PATRAS [3 TOTAL]: IST2001-32746-INSPIRE, PROMETHEUS-214901, 2023 ITEA-AMBIENCE

COMPUTER TECHNOLOGY INSTITUTE [1 TOTAL]: I-IMAS

LEADERSHIP, COORDINA- TION, AND POLICY- MAKING

Member of the Scientific Council of the Institute of Informatics and Telecommunications (IIT) from 2021 to 2025 (two terms), actively contributing to the Institute’s governance and strategic planning.

Head of the autonomous research group "MagCIL" (Multimedia Analysis Group, magcil.github.io) since 2020. The group comprises 10 members (2 PhD candidates and 8 collaborators) and conducts research in areas such as speech analysis, music information retrieval and organization, and multimedia analysis and generation with applications in health, culture, environment, and defense/security.

Member of the Research Exploitation Committee of NCSR “Demokritos” from 2021 to 2024, and Chair of the Committee in 2024. The committee is responsible for evaluating companies collaborating with the Technological Park and provides recommendations to the Center’s Board of Directors regarding research exploitation policy.

Scientific Coordinator of the two main postgraduate programs of the Institute of Informatics and Telecommunications ("Artificial Intelligence" and "Data Science"), and member of their Steering Committees and Joint Inter-Institutional Committees, actively involved in decision-making regarding curriculum design, program organization, and administration.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, Pascal, UNIX shell scripting, Matlab

Databases: mySQL, PostgreSQL, mongo

Web-development: PHP, JavaScript, JSP, ASP

Other packages: Pentaho Weka, SPSS, R, MS Office

Operating Systems: Linux, MacOS, Microsoft Windows family

PUBLICATIONS

BOOKS

- 1 Theodoros Giannakopoulos and Aggelos Pikrakis, Introduction to Audio Analysis: A MATLAB Approach, Academic Press, 2014, ISBN: 9780080993881

PATENTS

- 1 NCSR Demokritos, E. Spyrou, T. Giannakopoulos, S. Perantonis, D. Sgouropoulos, G. Siantikos, K. Spyropoulos, System for speaker discrimination based on audio-visual data, Industrial Property Organization (Greek), 1008860 - 27/09/2016

JOURNALS

- 18 Mitsou Alexandros, Petrogianni Antonia, Vakalaki Eleni Amvrosia, Nikou Christos, Theodoros Psallidas, and Theodoros Giannakopoulos, A multimodal dataset for electric guitar playing technique recognition, Data in Brief, Vol 52, 2024,
- 17 Alygizakis, N., Giannakopoulos, T., Thomaidis, N. S., Slobodnik, J. (2022). Detecting the sources of chemicals in the Black Sea using non-target screening and deep learning convolutional neural networks. Science of The Total Environment, 847, 157554.
- 16 Stoumpou, V., Vargas, C. D., Schade, P. F., Boyd, J. L., Giannakopoulos, T., Jarvis, E. D. (2022). Analysis of Mouse Vocal Communication (AMVOC): a deep, unsupervised method for rapid detection, analysis and classification of ultrasonic vocalisations. Bioacoustics, 1-31.
- 15 Koromilas, P., Giannakopoulos, T. (2021). Deep multimodal emotion recognition on human speech: A review. Applied Sciences, 11(17), 7962.
- 14 Psallidas, T., Koromilas, P., Giannakopoulos, T., Spyrou, E. (2021). Multimodal summarization of user-generated videos. Applied Sciences, 11(11), 5260.
- 13 Paraskevopoulos, G., Spyrou, E., Sgouropoulos, D., Giannakopoulos, T., Mylonas, P. (2019). Real-Time Arm Gesture Recognition Using 3D Skeleton Joint Data. Algorithms, 12(5), 108.
- 12 Papakostas, Michalis, and Theodoros Giannakopoulos. "Speech-Music Discrimination Using Deep Visual Feature Extractors." Expert Systems with Applications (2018)
- 11 Nikolaos Sarafianos, Theodoros Giannakopoulos, Christophoros Nikou, Ioannis A. Kakadiaris, Curriculum learning of visual attribute clusters for multi-task classification, Pattern Recognition, Volume 80, August 2018, Pages 94-108
- 10 Bougiatiotis, Konstantinos, and Theodoros Giannakopoulos. "Enhanced Movie Content Similarity Based on Textual, Auditory and Visual Information." Expert Systems with Applications (2017).
- 9 Papakostas, Michalis; Spyrou, Evaggelos; Giannakopoulos, Theodoros; Siantikos, Giorgos; Sgouropoulos, Dimitrios; Mylonas, Phivos; Makedon, Fillia, "Deep Visual Attributes vs. Hand-Crafted Audio Features on Multidomain Speech Emotion Recognition.", 2017 Computation 5, no. 2: 26
- 8 Nivolianitou, Z. S., Koromila, I. A., Giannakopoulos, T. "Bayesian network to predict environmental risk of a possible ship accident", International Journal of Risk Assessment and Management, 19(3), 228-239, 2016

- 7 Sarafianos, Nikolaos, Theodoros Giannakopoulos, and Sergios Petridis. "Audio-visual speaker diarization using fisher linear semi-discriminant analysis." *Multimedia Tools and Applications* 75.1 (2016): 115-130.
- 6 Giannakopoulos Theodoros. *pyAudioAnalysis: An Open-Source Python Library for Audio Signal Analysis*. PloS one 10.12 (2015): e0144610.
- 5 Petridis, S., Giannakopoulos, T., Spyropoulos, C. D. (2015). A Low Cost Pupilometry Approach. *International Journal of E-Health and Medical Communications (IJEHMC)*, 6(4), 49-61.
- 4 Theodoros Giannakopoulos and Sergios Petridis. Fisher Linear Semi-discriminant Analysis for Speaker Diarization, *IEEE Transactions On Audio, Speech, And Language Processing*, Vol. 20, No. 7, September 2012
- 3 Thanassis Perperis, Theodoros Giannakopoulos, Alexandros Makris, Dimitrios I. Kosmopoulos, Sofia Tsekeridou, Stavros J. Perantonis and Sergios Theodoridis. Multi-modal and ontology-based fusion approaches of audio and visual processing for violence detection in movies. *Expert Systems with Applications*, Vol 38(11), October 2011, pp: 14102-14116
- 2 Aggelos Pikrakis, Theodoros Giannakopoulos, and Sergios Theodoridis. A speech - music discriminator of radio recordings based on dynamic programming and bayesian networks. *Multimedia, IEEE Transactions on*, 10(5):846-857, 2008.
- 1 Theodoros Giannakopoulos, Nikolas Tatlas, Todor Ganchev, and Ilias Potamitis. A practical, real-time speech-driven home automation front-end. *IEEE Transactions on Consumer Electronics*, 51(2):514-523, 2005.

BOOK CHAPTERS

- 8 Touros, Georgios, and Theodoros Giannakopoulos. "Video Soundtrack Evaluation with Machine Learning: Data Availability, Feature Extraction, and Classification." *Advances in Speech and Music Technology: Computational Aspects and Applications*. Cham: Springer International Publishing, 2022. 137-157.
- 7 Geroulanos, Angelos, and Theodoros Giannakopoulos. "Emotion Recognition in Music Using Deep Neural Networks." *Advances in Speech and Music Technology: Computational Aspects and Applications*. Cham: Springer International Publishing, 2022. 193-213.
- 6 Giannakopoulos, T., Konstantopoulos, S., Siantikos, G., Karkaletsis, V. (2019). A System of Recognition Services for Clinical Assessment. In *RADIO-Robots in Assisted Living* (pp. 7-18). Springer
- 5 Sotirios Gyftakis, Ioanna Koromila, Theodore Giannakopoulos, Zoe Nivolianitou, Eleni Charou, Stavros Perantonis, Decision Support Tool Employing Bayesian Risk Framework for Environmentally Safe Shipping in Modeling, Computing and Data Handling Methodologies for Maritime Transportation pp 117-143, Part of the Intelligent Systems Reference Library book series (ISRL, volume 131)
- 4 D. Sgouropoulos, Th. Giannakopoulos, G. Siantikos, E. Spyrou and S. Perantonis, Detection of Clothes Change Fusing Color, Texture, Edge and Depth Information. M.S. Obaidat, A. Holzinger and J. Filipe (eds.), *E-Business and Telecommunications*, vol. 554 of Communications in Computer and Information Science, pp. 383-392, Springer, 2015

- 3 Giannakopoulos, Theodoros, Stasinos Konstantopoulos, Georgios Siantikos, and Vangelis Karkaletsis. "Design for a System of Multimodal Interconnected ADL Recognition Services." In *Components and Services for IoT Platforms*, pp. 323-333. Springer International Publishing, 2017
- 2 Petridis, Sergios, Theodoros Giannakopoulos, and Stavros Perantonis. "Unobtrusive Low-Cost Physiological Monitoring Using Visual Information." *Handbook of Research on Innovations in the Diagnosis and Treatment of Dementia* (2015): 306.
- 1 Aggelos Pikrakis, Theodoros Giannakopoulos, and Sergios Theodoridis. An overview of speech - music discrimination techniques in the context of audio recordings. *Multimedia Services in Intelligent Environments (Advanced Tools and Methodologies, Studies in Computational Intelligence)*, 120:81-102, 2008.

BOOK TRANSLATIONS - EDITING

- 2 Translation (to Greek) of the book: "Multimedia Systems: Algorithms, Standards, and Industry Practices", Parag Havaldar and Gerard Medioni
- 1 Translation (to Greek) and editing of the book: "Pattern Recognition", Sergios Theodoridis and Konstantinos Koutroumbas

CONFERENCES

- 79 Kaliosis, Panagiotis and Eleftheriou, Sofia and Nikou, Christos and Giannakopoulos, Theodoros, A self-supervised learning approach for detecting non-psychotic relapses using wearable-based digital phenotyping, 2024 IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (ICASSPW), pages=97-98,
- 78 Panagiotis Koromilas and Giorgos Bouritsas and Theodoros Giannakopoulos and Michalis Nicolaou and Yannis Panagakis, Bridging Mini-Batch and Asymptotic Analysis in Contrastive Learning: From InfoNCE to Kernel-Based Losses, Forty-first International Conference on Machine Learning, 2024
- 77 Evangelia Gkritzali, Panagiotis Kaliosis, Sofia Galanaki, Elisavet Palogiannidi and Theodoros Giannakopoulos Greek2MathTex: A Greek Speech-to-Text Framework for LaTeX Equations Generation, 13th Hellenic Conference on AI, SETN 2024, Piraeus, Greece, September 11-13, 2024. Proceedings 4
- 76 Antonia Petrogianni, Lefteris Kapelonis, Nikos Antoniou, Sofia Eleftheriou, Petros Mitseas, Dimitris Sgouropoulos, Nassos Katsamanis, Thodoris Giannakopoulos and Shrikanth Narayanan, RobuSER: A robustness Benchmark for Speech Emotion Recognition, 2024 12th International Conference on Affective Computing and Intelligent Interaction (ACII) (pp. 1-7). IEEE.
- 75 Dimitrios Sgouropoulos, Petros Mitseas, Sofia Eleftheriou, Theodoros Giannakopoulos, Antonia Petrogianni, Lefteris Kapelonis, Nikolaos Antoniou, Nassos Katsamanis and Shrikanth Narayanan, Emotion-Aware Speech Popularity Prediction: A Use-Case on Ted Talks, 2024 12th International Conference on Affective Computing and Intelligent Interaction (ACII) (pp. 1-7). IEEE, 2024
- 74 Bochalidis, C., Vargas, C. D., Jarvis, E. D., Giannakopoulos, T. (2023, August). Un-supervised Temporal Analysis of Mouse Vocalizations. In 2023 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB) (pp. 1-8). IEEE.

- 73 Christopoulos, D., Chatzi, E., Sofianopoulos, G., Patiniotaki, E., Eleftheriou, S., Koromilas, P., ... Sismanidou, E. (2023, September). Smart Subs Subtitling App for Watching Live Virtual Dome Performances. In 2023 18th International Workshop on Semantic and Social Media Adaptation Personalization (SMAP) 18th International Workshop on Semantic and Social Media Adaptation Personalization (SMAP 2023) (pp. 1-6). IEEE.
- 72 Koromilas, P., Nicolaou, M. A., Giannakopoulos, T., Panagakis, Y. (2023, June). MMATR: A Lightweight Approach for Multimodal Sentiment Analysis Based on Tensor Methods. In ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 1-5). IEEE.
- 71 Antoniou, N., Katsamanis, A., Giannakopoulos, T., Narayanan, S. (2023, June). Designing and Evaluating Speech Emotion Recognition Systems: A reality check case study with IEMOCAP. In ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 1-5). IEEE.
- 70 Papaioannou Charilaos, Valiantzas Ioannis, Giannakopoulos Theodoros, Kaliakatsos-Papakostas Maximos and Potamianos, Alexandros, A Dataset for Greek Traditional and Folk Music: Lyra, 23rd International Society for Music Information Retrieval Conference (ISMIR 2022)
- 69 Chatziagapi, A., Sgouropoulos, D., Karouzos, C., Melistas, T., Giannakopoulos, T., Katsamanis, A., Narayanan, S. (2022, October). Audio and ASR-based Filled Pause Detection. In 2022 10th International Conference on Affective Computing and Intelligent Interaction (ACII) (pp. 1-7). IEEE
- 68 Moutti, M., Eleftheriou, S., Koromilas, P., Giannakopoulos, T. (2022, June). Cross-linguistic speech emotion recognition using CNNs: a use-case in Greek Theatrical Data. In Proceedings of the 15th International Conference on Pervasive Technologies Related to Assistive Environments (pp. 662-667)
- 67 Tsitos, A. C., Dagioglou, M., Giannakopoulos, T. (2022, March). Real-time feasibility of a human intention method evaluated through a competitive human-robot reaching game. In 2022 17th ACM/IEEE International Conference on Human-Robot Interaction (HRI) (pp. 1080-1084). IEEE
- 66 Maria Moutti, Sofia Eleftheriou, Panagiotis Koromilas, Theodoros Giannakopoulos, A Dataset for Speech Emotion Recognition in Greek Theatrical Plays, Proceedings of the 13th Conference on Language Resources and Evaluation (LREC 2022) 1040–1046
- 65 Dagioglou, M., Soulounias, N., Giannakopoulos, T. (2022). Object Size Prediction from Hand Movement Using a Single RGB Sensor. In International Conference on Human-Computer Interaction (pp. 369-386). Springer, Cham
- 64 Petrogianni, A., Koromilas, P., Giannakopoulos, T. (2022). Film Shot Type Classification Based on Camera Movement Styles. In Iberian Conference on Pattern Recognition and Image Analysis (pp. 602-615). Springer, Cham.
- 63 Mitsou, A., Spyrou, E., Giannakopoulos, T. (2021, November). Multimodal Workplace Monitoring for Human Activity Recognition. In 25th Pan-Hellenic Conference on Informatics (pp. 206-211).
- 62 Paraskevoudis, K., Giannakopoulos, T. (2021, April). Instrument Playing Technique Recognition: A Greek Music Use Case. In Worldwide Music Conference (pp. 124-136). Springer, Cham.

- 61 Thomas Melistas, Theodoros Giannakopoulos, Lyrics and Vocal Melody Generation conditioned on Accompaniment Proceedings of the 2nd Workshop on NLP for Music and Spoken Audio (NLP4MusA, 2021)
- 60 Eleftheriou, S., Koromilas, P., Giannakopoulos, T. (2021). Automatic Assessment of Speaking Skills Using Aural and Textual Information. In Proceedings of The Fourth International Conference on Natural Language and Speech Processing (ICNLSP 2021) (pp. 166-175)
- 59 Psallidas, T., Mitsou, A., Pikramenos, G., Spyrou, E., Giannakopoulos, T. (2020, October). ARCHEO: A Dataset for Sound Event Detection in Areas of Touristic Interest. In 2020 15th International Workshop on Semantic and Social Media Adaptation and Personalization (SMA) (pp. 1-6). IEEE.
- 58 Giannakopoulos, T., Dimopoulos, S., Pantazopoulos, G., Chatziagapi, A., Sgouropoulos, D., Katsamanis, A., ... Narayanan, S. (2019, September). Using Oliver API for emotion-aware movie content characterization. In 2019 International Conference on Content-Based Multimedia Indexing (CBMI) (pp. 1-4). IEEE.
- 57 Chatziagapi, A., Paraskevopoulos, G., Sgouropoulos, D., Pantazopoulos, G., Nikandrou, M., Giannakopoulos, T., ... Narayanan, S. (2019). Data Augmentation Using GANs for Speech Emotion Recognition. Proc. Interspeech 2019, 171-175.
- 56 Paraskevopoulos, G., Tzinis, E., Ellinas, N., Giannakopoulos, T., Potamianos, A. (2019). Unsupervised Low-Rank Representations for Speech Emotion Recognition. Proc. Interspeech 2019, 939-943.
- 55 Giannakopoulos, T., Orfanidi, M., Perantonis, S. (2019, January). Athens Urban Soundscape (ATHUS): A Dataset for Urban Soundscape Quality Recognition. In International Conference on Multimedia Modeling (pp. 338-348). Springer, Cham.
- 54 Giannakopoulos T., Spyrou E, and Perantonis S. "Recognition of urban sound events using deep context-aware feature extractors and handcrafted features." IFIP International Conference on Artificial Intelligence Applications and Innovations. Springer, Cham, 2019.
- 53 Giannakopoulos T, and Perantonis S. "Recognizing the quality of urban sound recordings using hand-crafted and deep audio features." Proceedings of the 12th ACM International Conference on Pervasive Technologies Related to Assistive Environments. ACM, 2019.
- 52 Papakostas, M., Tsiakas, K., Giannakopoulos, T., Makedon, F. (2017, December). Towards predicting task performance from EEG signals. In Big Data (Big Data), 2017 IEEE International Conference on (pp. 4423-4425). IEEE.
- 51 Spyrou, E., Giannakopoulos, T., Sgouropoulos, D., and Papakostas, M. (2017, July). Extracting emotions from speech using a bag-of-visual-words approach. In Semantic and Social Media Adaptation and Personalization (SMAP), 2017 12th International Workshop on (pp. 80-83). IEEE.
- 50 N. Sarafianos, T. Giannakopoulos, C. Nikou, and I.A. Kakadiaris, Curriculum Learning for Multi-Task Classification of Visual Attributes, Proc. IEEE International Conference on Computer Vision Workshops 2017
- 49 Papakostas, M., Giannakopoulos, T., & Karkaletsis, V. (2017, June). A Fitness Monitoring System based on Fusion of Visual and Sensorial Information. In Proceedings of the 10th International Conference on Pervasive Technologies Related to Assistive Environments (pp. 280-285). ACM. ISO 690

- 48 Theodoros Giannakopoulos and Stasinos Konstantopoulos, Daily Activity Recognition based on Meta-classification of Low-level Audio Events, Proceedings of ICT4AWE 2017, ISBN: 978-989-758-251-6
- 47 Keramidas, G., Antonopoulos, C., Voros, N. S., Schwiegelshohn, F., Wehner, P., Rettkowski, J., Göhringer D., Hübner M., Konstantopoulos S., Giannakopoulos T., Karkaletsis, V. and Mariatos V. (2016, March). Computation and communication challenges to deploy robots in assisted living environments. In Design, Automation & Test in Europe Conference & Exhibition (DATE), 2016 (pp. 888-893). IEEE
- 46 Papakostas, M., Siantikos, G., Giannakopoulos, T., Spyrou, E., & Sgouropoulos, D. Recognizing Emotional States Using Speech Information., 2017 In GeNeDis 2016 (pp. 155-164). Springer, Cham
- 45 M.Papakostas, T. Giannakopoulos, F. Makedon, V. Karkaletsis, "Short-term Recognition of Human Activities using Convolutional Neural Networks" ,2016 International Conference on Signal-Image Technology & Internet Based Systems (SITIS). IEEE, 2016
- 44 Georgios Siantikos, Theodoros Giannakopoulos, Stasinos Konstantopoulos, A Low-cost Approach for Detecting Activities of Daily Living using Audio Information: A Use Case on Bathroom Activity Monitoring, 2nd International Conference on Information and Communication Technologies for Ageing Well and e-Health (ICT4AWE 2016), Rome, Italy, 21-22 April 2016
- 43 Theodoros Giannakopoulos, Georgios Siantikos, "A ROS Framework for Audio-Based Activity Recognition", In Proceedings of the 9th ACM International Conference on Pervasive Technologies Related to Assistive Environments (PETRA 2016)
- 42 Christos Smailis, Nikolaos Sarafianos, Theodoros Giannakopoulos, Stavros Perantonis, "Fusing Active Orientation Models and Mid-term Audio Features for Automatic Depression Estimation", In Proceedings of the 9th ACM International Conference on Pervasive Technologies Related to Assistive Environments (PETRA 2016)
- 41 Bougiatiotis, Konstantinos, and Theodoros Giannakopoulos. "Content Representation and Similarity of Movies based on Topic Extraction from Subtitles." Proceedings of the 9th Hellenic Conference on Artificial Intelligence. ACM, 2016.
- 40 Antonopoulos, Christos, Georgios Keramidas, Nikolaos S. Voros, Michael Hübner, Diana Göhringer, Maria Dagioglou, Theodore Giannakopoulos, Stasinos Konstantopoulos, and Vangelis Karkaletsis. "Robots in assisted living environments as an unobtrusive, efficient, reliable and modular solution for independent ageing: The RADIO perspective." In International Symposium on Applied Reconfigurable Computing, pp. 519-530. Springer International Publishing, 2015.
- 39 Charou, E., Bratsolis, E., Gyftakis, S., Giannakopoulos, T., & Perantonis, S. (2015, July). Use of Sentinel-1 data for Maritime Domain Awareness: Preliminary results. In Information, Intelligence, Systems and Applications (IISA), 2015 6th International Conference on (pp. 1-4). IEEE.
- 38 Sgouropoulos, D., Spyrou, E., Siantikos, G., & Giannakopoulos, T. (2015, November). Counting and tracking people in a smart room: An IoT approach. In Semantic and Social Media Adaptation and Personalization (SMAP), 2015 10th International Workshop on (pp. 1-5). IEEE.
- 37 Siantikos, G., Sgouropoulos, D., Giannakopoulos, T., Spyrou, E. (2015, September). Fusing multiple audio sensors for acoustic event detection. In Image and Signal Processing and Analysis (ISPA), 2015 9th International Symposium on (pp. 265-269). IEEE.

- 36 Giannakopoulos, T., Papakostas, M., Perantonis, S., Karkaletsis, V. (2015, September). Visual sentiment analysis for brand monitoring enhancement. In Image and Signal Processing and Analysis (ISPA), 2015 9th International Symposium on (pp. 1-6). IEEE.
- 35 Koromila, I., Nivolianitou, Z., Perantonis, S., Giannakopoulos, T., Charou, E., Gyftakis, S., Spyrou, K. (2015). Environmental Risk Assessment for the Aegean Sea. 11th International Conference on Marine Navigation and Safety of Sea Transportation (TransNav 2015)
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OTHER ACTIVITIES

2015 - 2018: CEO and Cofounder of the Ramon Private Company (www.ramon.gr)

Reviewer:

In conferences:

- International Conference on Digital Signal Processing (DSP): 2008, 2011, 2013
- International Conference on Pattern Recognition (ICPR): 2010, 2012, 2016

- Hellenic Conference on Artificial Intelligence (SETN): 2010, 2012
- European Signal Processing Conference (EUSIPCO): 2012
- IEEE Control Systems Society Conference (ACC): 2020

In journals:

- Transactions on Multimedia - IEEE
- Transactions on Transactions on Audio, Speech and Language Processing - IEEE
- Electronics Letters - IEEE
- PLOS ONE
- Artificial Intelligence Review - ELSEVIER
- Journal of Computational Intelligence - Wiley Online Library
- Pattern Recognition Letters - ELSEVIER
- Neurocomputing - ELSEVIER
- Journal on Advances in Signal Processing - EURASIP
- Language Resources and Evaluation - SPRINGER

2004-2008: Administrator of the web site of the International Society for Pediatric Neurosurgery ispneurosurgery.org.

2007 - 2009: Co-founder of the Spyros Sgouros - MIP: Medical Image Processing Co (reg: 246433), focusing on the development of web-based medical image processing tools.

2008 - 2011: Administrator of the web site of the European Association for Signal Processing - EURASIP: eurasip.org.

FOREIGN LANGUAGES

ENGLISH: First Certificate (Cambridge)

GERMAN: Elementary understanding